Adagio Institute, Inc.
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First-Round Questions to the Candidate

Paper 1

The Physics of the Instant: Analysis from Limits, Relativity, and Renormalization

Candidate:	Benjamin D. Summers
Degree:	Doctor of Philosophy in Mathematical Physics (Concentration in Stochastic Quantum Theory), by Published Works
Doctoral Examiner:	E. F. Boero Adagio Institute, Inc.
Independent evaluation:	Dr S. R. Ismail-Sutton, Durham University
Identifiers:	Q-P01-01 to Q-P01-10 (technical report); Q-PF-01, Q-PF-02, Q-PF-09 (portfolio-level evaluation)
Compiled:	14 October 2026

This document belongs to the examination record (03_examiner_questions). It lists the questions put to the candidate in the first round of the examination of Paper 1 and gives each a stable identifier, by which the Report, the Technical Annex and the correspondence can cite it.

1 Sources and identifiers

Technical report. Section 2 transcribes Section 7, "Questions to the author", of the Doctoral Examiner's technical report on Paper 1, `TR-P01-DOCv2-EX01-Rv1.pdf`. The report identifies the version examined as SSRN, DOI 10.2139/ssrn.6922380, 7 September 2026. It carries the evaluation date 4 September 2026, which was carried over from the other reports of the batch (Technical Annex, Section T.2(f), 06_final_examiner_report). The Doctoral Examiner sent it to the candidate on 11 September 2026 (COR-P01-03), and the candidate answered on 14 September 2026 (COR-P01-04). The corrected report of 29 September 2026, `TR-P01-DOCv2-EX01-Rv2.pdf` (COR-P01-11), is the one deposited in Exhibit I (02_technical_evaluations). The questions below are those of the report as first sent, which the candidate answered.

Identifiers. `Q-P01-`k is Question k of that report. The candidate's letters and the Technical Annex cite the questions by these numbers. The text is the report's, word for word; its section signs (§) refer to sections of the paper, and its finding numbers to findings of the report. Notes in small type are not part of the questions. They identify unresolved references, later corrections and related questions.

Independent evaluation. The independent evaluator's review of Paper 1 (`TR-P01-DOCv1-EX02-Rv1.pdf`, Exhibit J, 02_technical_evaluations) states its points as comments and poses no questions. It was sent to the candidate on 6 September 2026 and is filed as COR-P01-01, the candidate's

forward of that message; the candidate answered it on 7 September 2026 (COR-P01-02). It is not transcribed here.

Portfolio-level evaluation. Section 3 transcribes the questions of the portfolio-level technical evaluation (version 2, 4 September 2026; Exhibit H, Section 7, 02_technical_evaluations) that concern Paper 1. `Q-PF-`k is Question k of that evaluation and has the same identifier in all six documents; Q-PF-09 concerns all six papers and appears in each. The evaluation names the works W1–W6, and Wn is Paper n.

2 Questions of the technical report

Source: `TR-P01-DOCv2-EX01-Rv1.pdf`, Section 7, Questions 1–10.

Q-P01-01 Define dynamical completeness mathematically, separately for deterministic and for stochastic theories, and say which of the two your thesis concerns at each point of the argument.

Note. The Doctoral Examiner's letter of 26 September 2026 (COR-P01-07) quotes "well-posed evolution from admissible initial data" as the wording of Question 1. The phrase is in Finding 6 of the report; Question 1 does not contain it.

Q-P01-02 How do Feller-type processes — non-Markov, but with Chapman–Kolmogorov-composing two-time kernels — bear on your thesis? Your evidence is about CK composition and your conclusion is about completeness; state the step that connects them.

Q-P01-03 Restate the window claim of §2 with the hypothesis that §6 already supplies, and account for the case of a Markov process observed through a non-lumpable function of its state, whose CK defect does not vanish under refinement.

Q-P01-04 Supply a proof, or a precise reference, for the refinement limit under the conditioned hypothesis, with the conditions and if possible a rate. Note that Burke–Rosenblatt and Kemeny–Snell concern lumpings of the state space rather than functionals of a path segment, and address the existence of a defect rather than its behaviour under refinement.

Q-P01-05 Incorporate the two strengthenings of Finding **??**, and specify the sampling floor a protocol must report.

Note. "Finding **??**" is an unresolved cross-reference in the report as sent, and no finding of the report states two strengthenings. The candidate reported the defect on 14 September 2026 (COR-P01-04). The Doctoral Examiner's letter of 26 September 2026 (COR-P01-07) states that the reference was meant to carry two strengthenings of the Section 6 prediction (the one-sided form of the test, and a criterion separating classical memory from quantum indivisibility) together with the sampling floor, and supplies the sampling floor (Technical Annex, Section T.2(f)).

Q-P01-06 Give the licence for the inference from spatial to temporal bounds without invoking the time–energy uncertainty relation you disclaim.

Q-P01-07 How does the time-slice axiom affect your use of the algebraic formulation, and why should the singularity of point-localised fields bear on the completeness of Cauchy data?

Q-P01-08 Distinguish the renormalization scale, the regulator, the effective-theory validity scale and experimental resolution, and say which of them your argument requires.

Q-P01-09 Reconcile §7's claim to remain within confirmed physics with §3's statement that it extrapolates beyond it.

Q-P01-10 What empirical outcome would falsify the foundational thesis, as distinct from falsifying a chosen observation model?

3 Questions of the portfolio-level evaluation

Source: portfolio-level technical evaluation, version 2, 4 September 2026 (Exhibit H, 02_technical_evaluations), Section 7. Questions 1 and 2 concern Paper 1; Question 9 concerns all six papers.

Q-PF-01 **W1 alignment.** W2 §8 now concedes that $J(R) > 0$ does not distinguish indivisible dynamics from coarse-graining of a divisible process, and W6 proves that ordinary lumping produces CK violation with an exact balance law. Revise W1 §6 accordingly, or state what observable feature of two-time data on a finite grid does discriminate.

Q-PF-02 **W1, argument from limits.** Defend the premise that the instantaneous state is *constructed* as a limit of interval-averaged states, against the objection that in every formalism cited the point state is primitive and the averages derived. Alternatively withdraw the argument and develop the Haag–Kastler route, which is a theorem.

Q-PF-09 **Deposit venue.** State whether the works will be deposited on arXiv (math-ph / math.PR / quant-ph).